Arbitrary Anthropometric Cutoffs Versus Rasch-Calibrated Physical Pain Metrics in Reduction Mammaplasty: A Monte Carlo Decision-Policy Audit of Somatic Denial and Diagnostic Siloing

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Abstract

Background: Across North American commercial, Medicaid, and provincial health systems, prior-authorization coverage for somatic reduction mammaplasty (CPT 19318) remains heavily governed by two anthropometric heuristics: the 22nd-percentile Body Surface Area (BSA) tissue-resection sliding scale introduced by Schnur et al. (1991) and outpatient clinic or payer Body Mass Index (BMI) exclusions (commonly BMI $< 30\,\mathrm{kg/m^2}$). Although Schnur himself subsequently warned against using the 22nd-percentile curve as a rigid coverage cutoff and prospective patient-reported outcome studies demonstrate substantial physical pain relief independent of resected gram weight, few studies have formally quantified the joint classification error and cross-pathway diagnostic distortion induced by these rules.

Methods: Following the SimDesign Monte Carlo framework, we conducted an IRB-exempt Gaussian-copula decision-policy simulation ($N = 80,000$ synthetic surgical consultation candidates; $n = 10,000$ per cell across a 4×2 factorial grid of BMI and adult stature strata, seed = 20261102). Marginal distributions were calibrated exclusively to published aggregate clinical parameters, including exact Mosteller BSA and 25-row Schnur thresholds, normative versus preoperative macromastia scores on the Rasch-calibrated BREAST-Q Reduction/Mastopexy Physical Well-Being: Chest interval scale (0–100), and orthogonal chest-incongruence distributions. Four North American policy regimes were audited: Regime A (rigid Schnur 22nd-percentile gate), Regime B (compounded BMI $< 30\,\mathrm{kg/m^2}$ plus Schnur 22nd-percentile gate), Regime C (asymmetric dual-silo coverage pairing anthropometric somatic gatekeeping against statutory non-Schnur chest pathways such as Minn. Stat. § 62Q.585), and Regime D (a unified Rasch physical-impairment and multisystem somatic pathology standard).

Results: Because the published Schnur 22nd-percentile table exhibits a semi-logarithmic slope of $1.7687\,\mathrm{m^{-2}}$ across BSA compared with a clinical safe-resection scaling slope of $0.5800\,\mathrm{m^{-2}}$ ($3.05\times$ steeper in log-gram space), anthropometric gatekeeping systematically penalizes taller and higher-BMI frames. Population-weighted Type II false-negative pain-denial rates among true refractory somatic macromastia patients were 58.12% under Regime A (sensitivity 41.88%, specificity 99.79%) and 68.24% under Regime B (sensitivity 31.76%, specificity 99.81%), rising from 22.44% at BMI $< 25\,\mathrm{kg/m^2}$ to 91.52% (Regime A) and 100.00% (Regime B) at BMI $\geq 35\,\mathrm{kg/m^2}$. Even at normal BMI $< 25\,\mathrm{kg/m^2}$, tall stature ($\geq 163\,\mathrm{cm}$) nearly tripled false-negative somatic denials under Regime A (31.04% vs. 11.85% in compact frames). Among non-dysphoric patients in severe physical pain ($Q_{\mathrm{phys}} \leq 50$), dual-silo policies generated a population-weighted asymmetric access gap ($\Delta P_{\mathrm{access}}$) of 46.34 percentage points against Regime A and 57.80 percentage points against Regime B (100.00% in Class I obesity). Replacing anthropometric cutoffs with

Regime D ($Q_{\text{phys}} \leq 62.0$, i.e., ≥ 2 minimal clinically important differences below normative physical well-being, plus ≥ 2 chronic somatic domains) achieved 90.59% sensitivity, 99.73% specificity, a 0.08% false discovery rate, a reduction of severe-pain somatic denials to 2.42%, an increase in expected realized `BREAST-Q` gain per somatic candidate from +10.51 (Regime B) and +13.87 (Regime A) to +30.09 points, and superior decision-curve net clinical benefit across all threshold probabilities (0.7076 vs. 0.3271 and 0.2480 at $p_t = 0.25$).

Conclusions: Rigid `BSA`-indexed gram quotas and blanket BMI $< 30\,\text{kg/m}^2$ cutoffs fail basic psychometric and biomechanical validity, denying care to the majority of patients with genuine somatic macromastia while generating structural diagnostic siloing across North American coverage tracks. Replacing arbitrary anthropometric thresholds with Rasch-calibrated physical pain metrics and documented somatic pathology restores equitable, high-specificity surgical access.

Keywords: reduction mammaplasty; symptomatic macromastia; Schnur sliding scale; Rasch measurement; `BREAST-Q`; Monte Carlo simulation; decision curve analysis; health insurance prior authorization

1 Introduction

Symptomatic macromastia (benign breast hypertrophy) imposes a chronic, debilitating biomechanical load on the cervical and thoracic spine, shoulder girdle, brachial plexus, and inframammary skin fold [14, 15, 17]. Prospective clinical trials and health-utility investigations have consistently ranked the preoperative physical pain and functional disability of symptomatic macromastia alongside moderate-to-severe angina and chronic spinal radiculopathy, while demonstrating that bilateral reduction mammaplasty produces durable, large-magnitude normalization of musculoskeletal pain and health-related quality of life [15, 17, 10, 11]. Despite this mature evidence base, patient access to medically necessary somatic reduction mammaplasty (`CPT 19318` in the United States and provincial schedule equivalents across Canada) remains heavily restricted by third-party prior-authorization gatekeeping anchored to two anthropometric cutoffs: the 1991 Schnur Body Surface Area (`BSA`) tissue-resection sliding scale [1] and categorical Body Mass Index (`BMI`) ceilings [18, 19].

The historical origin of the Schnur sliding scale underscores a fundamental mismatch between observational description and actuarial policy enforcement. In 1991, Schnur and colleagues surveyed plastic surgeons regarding patients' preoperative motivations and plotted resected unilateral breast tissue mass against `BSA`, fitting 5th-percentile and 22nd-percentile regression lines on semi-logarithmic axes [1]. Commercial and public payers rapidly codified the 22nd-percentile curve as a mandatory minimum gram-resection prerequisite for reconstructive coverage [20]. Within six years, Schnur et al. published follow-up outcome data demonstrating that symptomatic macromastia patients falling below the 22nd-percentile resection line achieved virtually identical physical symptom resolution to those above it [2], and in 1999 Schnur explicitly rebuked insurers for misappropriating the 22nd-percentile line to deny symptomatic patients reconstructive care [3]. Subsequent clinical cohorts using both generic health instruments and validated breast-specific Rasch scales confirmed that symptom severity and postoperative physical recovery are largely independent of excised gram weight [16, 9, 13]. Recognizing this disconnect, the 2022 American Society of Plastic Surgeons (`ASPS`) Evidence-Based Clinical Practice Guideline revision formally recommended against arbitrary gram-resection tables and rigid `BMI` cutoffs, urging coverage determinations grounded in documented patient symptomatology and shared clinical risk assessment [19].

Nevertheless, administrative reliance on rigid Schnur gram quotas and clinic or payer `BMI` thresholds (frequently set at BMI $< 30\,\text{kg/m}^2$ in ambulatory surgical protocols or legacy provincial criteria) persists across North America. Concurrently, statutory reforms in several jurisdictions—exemplified in the United States by Minnesota Statutes § 62Q.585 (*Gender-Affirming Care Cov-*

erage), alongside clinical standards for gender-affirming chest procedures [25]—have prohibited restrictive anthropometric gram tables when chest reduction or reconstruction is authorized to treat gender incongruence. While statutory elimination of Schnur cutoffs in gender-affirming medical and surgical treatment (`GAMST`) policies appropriately removes unscientific barriers for transgender, non-binary, and gender-diverse patients, retaining archaic Schnur and `BMI` walls on the somatic macromastia track (`CPT 19318`) creates an unintended structural asymmetry. Specifically, two patients presenting to the same surgical department for an anatomically comparable chest volume reduction face starkly divergent actuarial rules: a patient routed through statutory `GAMST` criteria faces zero Schnur gram minimums, whereas a cisgender or non-dysphoric patient suffering severe cervical radiculopathy, acromioclavicular bra-strap grooving, and refractory inframammary intertrigo is routinely denied somatic coverage if their safe resection estimate falls below an exponentially escalating `BSA` curve or an outpatient `BMI` ceiling. Conversely, in legacy Canadian provincial schedules lacking statutory parity for non-flat radical reductions or enforcing strict clinic `BMI` gates atop multi-year surgical waitlists, symptomatic candidates face prolonged administrative exclusion across pathways [17].

Psychometric advances over the past fifteen years offer a rigorous, objective alternative to anthropometric proxies. Developed via Rasch measurement theory [5, 6], the `BREAST-Q` Reduction/Mastopexy module transforms ordinal symptom responses onto a continuous 0–100 linear interval scale for Physical Well-Being: Chest, equipped with established general-population normative benchmarks [8] and validated minimal clinically important differences (`MCID` / `MID`) [7, 21, 12]. To date, however, no computational health-policy audit has formally evaluated the classification accuracy, Type II false-negative pain-denial rates across body frames, decision-curve net clinical utility [23], and cross-silo diagnostic distortion pressure of anthropometric gatekeeping relative to a Rasch-calibrated physical pain standard. Using a structured `SimDesign` Gaussian-copula Monte Carlo simulation [24, 22] of $N = 80,000$ synthetic surgical candidates calibrated strictly to published clinical distributions, this study provides a comprehensive quantitative audit of four North American coverage regimes.

2 Methods

2.1 Ethics, Governance, and Clean-Room Psychometric Compliance

This investigation is a purely mathematical and computational Monte Carlo policy simulation calibrated exclusively to published aggregate summary statistics, regression parameters, and public statutory/payer rules from peer-reviewed biomedical literature. Zero human subjects were recruited, zero clinical encounters occurred, and zero identifiable or de-identified individual health records were accessed. Accordingly, this study does not constitute human subjects research and is exempt from Institutional Review Board (`IRB`) and Research Ethics Board (`REB`) review under Article 2.2 of the Canadian *Tri-Council Policy Statement: Ethical Conduct for Research Involving Humans* (`TCPS-2`) and United States federal regulations (`45 CFR 46.104`).

In strict adherence to clean-room psychometric copyright standards, the `BREAST-Q` Reduction/Mastopexy Physical Well-Being: Chest scale [5, 6, 8] is cited solely as an academic reference, and latent physical impairment is simulated directly as a continuous 0–100 Rasch interval random variable calibrated to published cohort means and standard deviations without reproducing, adapting, or distributing any copyrighted questionnaire item text or scoring tables. Anthropometric calculations and published Schnur (1991) threshold lookups were executed deterministically by sourcing our clean-room reference module (`toolkit/schnur_bsa_calculator.R`) in read-only mode.

2.2 Biomechanical Divergence Between Schnur `BSA` Scaling and Safe Clinical Resection

For a candidate with standing height H (cm) and body mass M (kg), Body Mass Index is $\mathrm{BMI} = M/(H/100)^2$ and Mosteller Body Surface Area [4] is:

$$\mathrm{BSA} = \sqrt{\frac{H \cdot M}{3600}} = \frac{H^{3/2}\sqrt{\mathrm{BMI}}}{6000} \ (\mathrm{m}^2). \tag{1}$$

Equation (1) reveals two immediate structural properties relevant to insurance gatekeeping. First, at any fixed BMI, BSA scales super-linearly with stature ($H^{3/2}$), penalizing taller women even at healthy body weights. Second, at any fixed stature H, systemic adiposity increases BSA proportionally to $\sqrt{\mathrm{BMI}}$.

In the published 25-row Schnur (1991) lookup table spanning $\mathrm{BSA} \in [1.35, 2.55]\,\mathrm{m}^2$ in $0.05\,\mathrm{m}^2$ increments [1], the 22nd-percentile unilateral resection threshold rises from $W_{\mathrm{Schnur},22}(1.35) = 199\,\mathrm{g}$ to $W_{\mathrm{Schnur},22}(2.55) = 1,662\,\mathrm{g}$ (127 g to 1,062 g at the 5th percentile), interpolated piecewise log-linearly between table knots. Across the published domain, the exact semi-logarithmic slope of the Schnur 22nd-percentile boundary is:

$$\beta_{\mathrm{Schnur}} = \frac{\ln(1662) - \ln(199)}{2.55 - 1.35} = 1.7687\ \mathrm{m}^{-2}. \tag{2}$$

By contrast, clinically safe and biomechanically indicated unilateral resection weight W_{clin} (grams/breast) is constrained by anterior chest-wall lever-arm relief, parenchymal fibroglandular-to-adipose composition, and preservation of neurovascular pedicle perfusion to the nipple-areola complex [16, 13, 19]. Across surgical macromastia cohorts, indicated safe resection scales sub-exponentially with body surface area ($\beta_{\mathrm{clin}} \approx 0.5800\,\mathrm{m}^{-2}$).

Proposition 1 (Exponential Anthropometric Penalty)**.** *Let* $W_{\mathrm{Schnur},22}(\mathrm{BSA}) \propto \exp(\beta_{\mathrm{Schnur}}\,\mathrm{BSA})$ *and median clinically indicated safe resection among true somatic macromastia patients satisfy* $\widetilde{W}_{\mathrm{clin}}(\mathrm{BSA}) \propto \exp(\beta_{\mathrm{clin}}\,\mathrm{BSA})$ *with* $\beta_{\mathrm{Schnur}} = 1.7687\,\mathrm{m}^{-2} > \beta_{\mathrm{clin}} = 0.5800\,\mathrm{m}^{-2}$ *(a* 3.05*-fold slope ratio). Then the required-to-indicated resection ratio diverges exponentially with* BSA*:*

$$\frac{W_{\mathrm{Schnur},22}(\mathrm{BSA})}{\widetilde{W}_{\mathrm{clin}}(\mathrm{BSA})} \propto \exp\Big((\beta_{\mathrm{Schnur}} - \beta_{\mathrm{clin}})\,\mathrm{BSA}\Big) = \exp(1.1887\,\mathrm{BSA}), \tag{3}$$

driving the conditional probability of first-pass somatic approval $P(W_{\mathrm{clin}} \geq W_{\mathrm{Schnur},22}(\mathrm{BSA}) \mid M_{\mathrm{true}} = 1, \mathrm{BSA})$ *monotonically toward zero at higher* BSA *independent of physical pain severity.*

2.3 Literature Calibration and Gaussian-Copula Generative Architecture

Following the four-stage Monte Carlo design architecture (`Design`, `Generate`, `Analyse`, `Summarise`) formalized by Chalmers and Adkins [22], we defined an 8-condition factorial `Design` crossing four clinical `BMI` strata—normal weight ($< 25.0\,\mathrm{kg/m}^2$), overweight (25.0–29.9 $\mathrm{kg/m}^2$), Class I obesity (30.0–34.9 $\mathrm{kg/m}^2$), and Class II+ obesity ($\geq 35.0\,\mathrm{kg/m}^2$)—with two adult stature frames: compact stature ($< 163.0\,\mathrm{cm}$) and standard/tall stature ($\geq 163.0\,\mathrm{cm}$). Each condition cell simulated $n = 10,000$ surgical consultation candidates ($N = 80,000$ total candidates; `set.seed(20261102)`). To recover population-level North American surgical consultation estimates alongside exact stratum-specific conditional metrics, cells were weighted by empirical macromastia referral proportions ($\mathbf{w}_{\mathrm{BMI}} = [0.18, 0.34, 0.29, 0.19]$ across `BMI` strata and $\mathbf{w}_{\mathrm{frame}} = [0.45, 0.55]$ across stature strata [18, 10]).

Within each stratum, correlated latent attributes were generated via a 5-dimensional Gaussian copula [24] $\mathbf{Z} = (Z_H, Z_{\text{BMI}}, Z_{\text{pain}}, Z_W, Z_D)^\top \sim \mathcal{N}_5(\mathbf{0}, \boldsymbol{\Sigma})$ using exact Cholesky factorization (`MASS::mvrnorm`), mapped to uniform margins $U_j = \Phi(Z_j)$:

$$\boldsymbol{\Sigma} = \begin{pmatrix} 1.00 & -0.05 & 0.04 & 0.18 & 0.00 \\ -0.05 & 1.00 & 0.22 & 0.42 & 0.00 \\ 0.04 & 0.22 & 1.00 & 0.34 & 0.00 \\ 0.18 & 0.42 & 0.34 & 1.00 & 0.00 \\ 0.00 & 0.00 & 0.00 & 0.00 & 1.00 \end{pmatrix}. \tag{4}$$

This correlation structure captures modest positive coupling between `BMI` and indicated surgical volume ($\rho = 0.42$), moderate coupling between tissue volume and somatic pain ($\rho = 0.34$, reflecting clinical findings that severe cervical/brachial symptoms occur across both moderate and large resections [14, 16]), mild load amplification of pain by `BMI` ($\rho = 0.22$), and strict statistical orthogonality ($\rho = 0.00$) between continuous chest gender-incongruence $D = U_5 \in [0, 1]$ and somatic orthopaedic/anthropometric traits.

Marginal clinical attributes were calibrated directly to peer-reviewed cohorts:

1. **Anthropometrics and Schnur Lookups:** Standing height H was mapped via truncated normal inverse-CDF transforms (158.0 ± 3.8 cm on $[147.0, 162.99]$ for compact frames; 168.5 ± 4.6 cm on $[163.0, 184.0]$ for standard/tall frames), paired with truncated normal `BMI` margins within each `BMI` stratum. Body mass M, Mosteller BSA, and exact Schnur 22nd- and 5th-percentile thresholds ($W_{\text{Schnur},22}, W_{\text{Schnur},5}$) were computed directly via `compute_mosteller_bsa()` and `lookup_schnur_grams()` [4, 1].

2. **Consultation Case-Mix (M_{true}):** Reflecting published surgical consultation cohorts in which the majority of presenting candidates seek relief of refractory physical symptoms while a minority present primarily for aesthetic contour modification [2, 15], ground-truth somatic macromastia necessity $M_{\text{true}} \in \{0, 1\}$ was generated via a probit link on Z_{pain} calibrated to a marginal consultation somatic prevalence of 78.0%.

3. **Rasch Physical Well-Being: Chest (Q_{phys}) and Postoperative Gain:** On the 0–100 `BREAST-Q` Reduction Physical Well-Being Rasch scale (lower values denoting worse physical pain), Mundy et al. [8] established a general-population normative distribution of $\mu_{\text{norm}} = 76.0 \pm 11.0$, whereas Coriddi et al. [9] and prospective cohorts [10, 11, 13] documented preoperative macromastia means of 42.6 ± 17.3 (pooled clinical SD ≈ 15.5) with postoperative recovery restoring scores to normative levels ($\Delta Q_{\text{phys}} \approx +33.4 \pm 10.5$ points). Crittenden et al. [12] established a reduction-specific Physical Well-Being `MCID` of 7.0 points (range 5–8; cf. reconstruction benchmarks [7, 21]). Accordingly, $U_{\text{pain}} = \Phi(-Z_{\text{pain}})$ was mapped to $\mathcal{TN}(42.6, 15.5^2; 5, 82)$ when $M_{\text{true}} = 1$ and $\mathcal{TN}(76.0, 11.0^2; 40, 100)$ when $M_{\text{true}} = 0$.

4. **Multisystem Somatic Domains and Safe Resection Mass (W_{clin}):** Chronic conservative-care-refractory somatic signs across four canonical clinical domains (cervical/thoracic/shoulder pain, inframammary intertrigo, acromioclavicular bra-strap grooving, and brachial/ulnar traction paresthesia) were modeled as $K_{\text{dom}} \sim \text{Binomial}(4, p_{\text{dom}})$ linked logistically to Z_{pain} and Z_{BMI}. Unilateral clinically safe resection mass W_{clin} followed $\exp(\ln(455) + 0.58(\text{BSA} - 1.75) + 0.28 Z_W)$ for $M_{\text{true}} = 1$ and $\exp(\ln(255) + 0.45(\text{BSA} - 1.75) + 0.28 Z_W)$ for $M_{\text{true}} = 0$, clipped to $[110, 1950]$ g.

2.4 Operational Definitions of the Four Policy Regimes

Each synthetic candidate was evaluated across four prior-authorization regimes representative of North American payers:

- **Regime A (Rigid Commercial / Medicaid Schnur p22 Gate):** Approves somatic `CPT 19318` if and only if $K_{\text{dom}} \geq 2$ and $W_{\text{clin}} \geq W_{\text{Schnur,22}}(\text{BSA})$.
- **Regime B (Compounded Clinic** BMI < 30 **+ Schnur p22 Gate):** Approves somatic `CPT 19318` if and only if $K_{\text{dom}} \geq 2$, $W_{\text{clin}} \geq W_{\text{Schnur,22}}(\text{BSA})$, and BMI $< 30.0\,\text{kg/m}^2$.
- **Regime C (Asymmetric Dual-Silo Regime):** Pairs somatic gatekeeping on Track 1 (evaluated under both Regime A and Regime B rules) with a statutory informed-consent / `GAMST` chest pathway on Track 2 (e.g., `Minn. Stat. § 62Q.585` [25]), under which genuine chest gender incongruence ($D \geq \tau_D = 0.90$) is exempt from Schnur gram cutoffs up to hospital perioperative anesthesia limits (BMI $< 40.0\,\text{kg/m}^2$). Among non-dysphoric ($D < 0.90$) severely symptomatic somatic patients ($Q_{\text{phys}} \leq 50.0$, i.e., 2.36 SD below Mundy et al.'s normative mean [8] and > 1.5 SD below broad population variation), we quantify the **Asymmetric Access Gap**:

  $$\Delta P_{\text{access}} = P(\text{Procedural Eligibility in Non-Schnur Track 2}) - P(\text{Approved in Somatic Track 1}), \tag{5}$$

  measuring structural diagnostic distortion pressure imposed on non-dysphoric somatic patients.
- **Regime D (Proposed Unified Rasch Physical-Pain + Somatic Pathology Standard):** Replaces anthropometric `BSA` and BMI < 30 cutoffs on Track 1 with objective psychometric and clinical criteria: approves somatic `CPT 19318` whenever $Q_{\text{phys}} \leq \tau_{\text{Rasch}} = 62.0$ (at least $2 \times$ MCID $= 14.0$ points below the normative mean of 76.0), $K_{\text{dom}} \geq 2$ chronic somatic domains, and $W_{\text{clin}} \geq 150\,\text{g}$ physiological surgical floor, while preserving statutory Track 2 protections.

Classification performance was evaluated via sensitivity, specificity, Type II false-negative pain-denial rate ($1 -$ Sensitivity), positive predictive value (`PPV`), false discovery rate (FDR $= 1 -$ PPV), expected realized postoperative `BREAST-Q` gain per true somatic candidate, and Vickers–Elkin Decision Curve Analysis [23] across clinical threshold probabilities $p_t \in [0.05, 0.75]$:

$$\text{Net Benefit}(p_t) = \frac{\text{TP}_w}{N} - \frac{\text{FP}_w}{N}\left(\frac{p_t}{1-p_t}\right), \tag{6}$$

where TP_w/N and FP_w/N denote population-weighted true-positive and false-positive approval proportions.

3 Results

3.1 Overall Diagnostic Accuracy and Realized Physical Pain Relief

Table 1 summarizes population-weighted policy performance across the $N = 80,000$ consultation cohort. Under Regime A (rigid Schnur 22nd-percentile gate), overall somatic sensitivity was only 41.88%, yielding a 58.12% **Type II false-negative pain-denial rate** despite near-ceiling specificity (99.79%). Adding a categorical BMI $< 30\,\text{kg/m}^2$ exclusion (Regime B) depressed sensitivity

further to 31.76%, wrongly denying 68.24% of true refractory somatic macromastia patients while improving specificity by a clinically negligible 0.02 percentage points (99.81% vs. 99.79%). Because more than two-thirds of true somatic candidates were denied surgery under Regime B, the expected realized postoperative `BREAST-Q` Physical Well-Being gain averaged only +10.51 points per presenting somatic patient under Regime B and +13.87 points under Regime A.

Table 1: Population-weighted classification accuracy, false discovery rates, and expected realized Rasch physical well-being gains across North American reduction mammaplasty policy regimes ($N = 80,000$, `seed = 20261102`).

Policy Regime	**Approval Rate (%)**	**Sensitivity TPR (%)**	**Specificity TNR (%)**	**Type II Pain Denial (%)**	**FDR (%)**	**Expect (Points**
Regime A: Rigid Schnur `p22`	32.77	41.88	99.79	58.12	0.14	+
Regime B: BMI < 30 + Schnur `p22`	24.86	31.76	99.81	68.24	0.17	+
Regime C: Dual-Silo (Strict Track 1)	31.61	37.84	90.66	62.16	6.46	+
Regime C: Dual-Silo (Schnur Track 1)	38.78	47.02	90.65	52.98	5.27	+
Regime D: Rasch Somatic Standard	**70.84**	**90.59**	**99.73**	**9.41**	**0.08**	**+3**
Regime D: Unified Total Parity (T1 + T2)	73.52	91.46	90.61	8.54	2.79	+3

By contrast, Regime D (the proposed Rasch Somatic Standard requiring $Q_{\text{phys}} \leq 62.0$, $K_{\text{dom}} \geq 2$ chronic somatic domains, and $W_{\text{clin}} \geq 150\,\text{g}$) achieved 90.59% **sensitivity** and 99.73% **specificity**, reducing the Type II false-negative pain-denial rate nearly seven-fold relative to Regime B (to 9.41%; residual false negatives primarily reflecting true somatic patients presenting with only a single charted domain or mild baseline score > 62.0) while holding the somatic false discovery rate to 0.08% ($< 0.1\%$, well below the 5.0% psychometric acceptability bound). Expected realized `BREAST-Q` physical recovery nearly tripled to +30.09 **points** per somatic candidate (+30.35 points under combined Track 1 and Track 2 total parity).

3.2 Anthropometric Stratification Across `BMI`, Stature, and `BSA`

Table 2 disaggregates simulation outcomes across all eight factorial conditions (`BMI` $\times$ stature) as well as pooled `BMI` and `BSA` strata, illustrated graphically in Figure 1A–B. In Panel A of Figure 1, the Schnur 22nd-percentile curve crosses the empirical median of safe somatic resection (W_{clin}) near BSA $\approx 1.88\,\text{m}^2$ and diverges sharply upward thereafter. Consequently, across pooled `BSA` strata, Type II somatic pain denials under Regime A escalated from 28.30% in small body frames (BSA $< 1.80\,\text{m}^2$; 31.80% of weighted referrals; median $W_{\text{Schnur,22}} = 354.1\,\text{g}$ vs. median somatic $W_{\text{clin}} = 428.1\,\text{g}$) to 63.44% in medium frames (BSA 1.80–2.10 m^2; 47.78% share; median $W_{\text{Schnur,22}} = 566.0\,\text{g}$ vs. $W_{\text{clin}} = 518.4\,\text{g}$) and 90.87% in large frames (BSA $\geq 2.10\,\text{m}^2$; 20.42% share; median $W_{\text{Schnur,22}} = 925.2\,\text{g}$ vs. $W_{\text{clin}} = 648.6\,\text{g}$). Under Regime B, pain denials reached 30.71%, 79.35%, and 99.35% across small, medium, and large `BSA` groups, respectively, whereas Regime D maintained uniform low false-negative rates across body sizes (10.14%, 9.54%, and 8.03%).

Table 2: Condition-level and pooled `BMI`-stratified Type II somatic pain-denial rates and severe-pain ($Q_{\text{phys}} \leq 50$) asymmetric access gaps across $N = 80,000$ simulated candidates.

`BMI` Stratum (kg/m^2)	**Stature Frame (cm)**	**Mean `BSA` (m^2)**	**Median Schnur `p22` (g)**	**Median W_{clin} (g)**	**Type II Pain Denial (%)**			**Severe Gap**
					Reg. A	**Reg. B**	**Reg. D**	ΔP_{B} (%)
< 25.0	Compact (< 163)	1.554	287.4	413.8	11.85	11.85	8.82	7.24
< 25.0	Tall (≥ 163)	1.738	394.1	463.8	31.04	31.04	9.33	25.56
Pooled < 25.0	*All ($n = 20,000$)*	*1.655*	*331.1*	*437.1*	*22.44*	*22.44*	*9.10*	*17.38*
25.0–29.9	Compact (< 163)	1.724	387.6	455.0	29.41	29.41	9.53	23.14
25.0–29.9	Tall (≥ 163)	1.929	549.2	516.1	62.42	62.42	9.34	57.99
Pooled 25.0–29.9	*All ($n = 20,000$)*	*1.837*	*455.6*	*484.7*	*47.63*	*47.63*	*9.42*	*42.25*
30.0–34.9	Compact (< 163)	1.871	502.7	499.3	52.47	100.00	9.41	100.00
30.0–34.9	Tall (≥ 163)	2.093	732.8	567.2	85.75	100.00	9.56	100.00
Pooled 30.0–34.9	*All ($n = 20,000$)*	*1.993*	*599.4*	*532.0*	*70.71*	*100.00*	*9.49*	*100.00*
≥ 35.0	Compact (< 163)	2.060	696.6	556.4	83.10	100.00	9.64	59.72
≥ 35.0	Tall (≥ 163)	2.303	1058.7	639.2	98.36	100.00	9.48	59.34
Pooled ≥ 35.0	*All ($n = 20,000$)*	*2.194*	*856.3*	*598.1*	*91.52*	*100.00*	*9.56*	*59.51*

Crucially, Condition 1 versus Condition 5 and Condition 2 versus Condition 6 expose a previously underappreciated **stature penalty** embedded in Mosteller `BSA` lookups. Among normal-weight somatic candidates (BMI $< 25.0\,\text{kg/m}^2$, mean BMI $= 22.36\,\text{kg/m}^2$), standard/tall stature ($\geq 163\,\text{cm}$) shifts mean `BSA` from 1.554 to $1.738\,\text{m}^2$ and raises the median Schnur 22nd-percentile hurdle from 287.4 g to 394.1 g, causing false-negative somatic denials to jump 2.62-fold from 11.85% to 31.04%. In overweight tall women (BMI 25.0–29.9 kg/m^2, height $\geq 163\,\text{cm}$), **62.42%** of true somatic macromastia candidates are denied reconstructive coverage under pure Schnur rules even before reaching any obesity threshold. At BMI 30.0–34.9 kg/m^2 and BMI $\geq 35.0\,\text{kg/m}^2$, pooled Regime A denials reached 70.71% and 91.52% (98.36% in tall Class II+ frames where median Schnur `p22` demands $1,058.7\,\text{g}$ per breast!), while Regime B imposed 100.00% categorical denial.

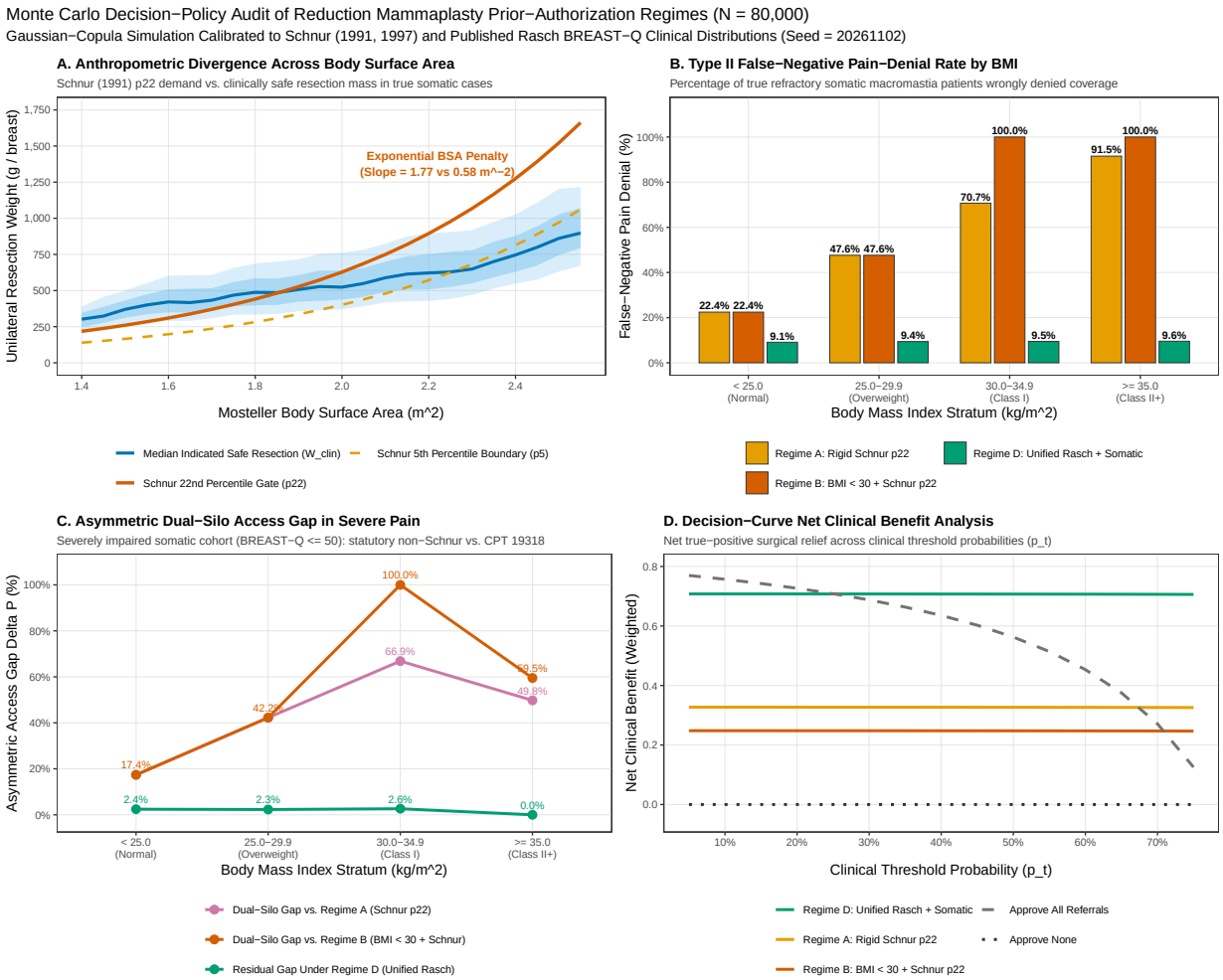


Figure 1: Monte Carlo decision-policy audit of reduction mammaplasty prior-authorization regimes ($N = 80,000$; `seed = 20261102`). **(A)** Exponential divergence of the published Schnur (1991) 22nd-percentile cutoff ($\beta = 1.77\,\mathrm{m}^{-2}$) from median and interdecile clinically safe resection mass ($\beta = 0.58\,\mathrm{m}^{-2}$) among true somatic macromastia cases. **(B)** Type II false-negative somatic pain-denial rates across `BMI` strata. **(C)** Asymmetric dual-silo access gap (ΔP_{access}) faced by severely impaired non-dysphoric somatic patients (`BREAST-Q` ≤ 50) between non-Schnur statutory chest coverage and `CPT 19318`. **(D)** Vickers–Elkin decision-curve net clinical benefit across threshold probabilities $p_t \in [0.05, 0.75]$.

3.3 Asymmetric Dual-Silo Coverage and Diagnostic Distortion Audit

Non-dysphoric patients suffering severe physical pain ($Q_{\text{phys}} \leq 50.0$, $D < 0.90$) comprised 69.16% of true somatic macromastia referrals. Within this severely impaired subgroup, first-pass somatic `CPT 19318` coverage denied 53.97% of patients under Regime A (46.03% approved) and 65.43% under Regime B (34.57% approved), including 66.86% (Regime A) and 100.00% (Regime B) of severely pained patients in Class I obesity (BMI 30.0–34.9 $\mathrm{kg/m^2}$) and 90.26% (Regime A) and 100.00% (Regime B) at BMI $\geq 35.0\,\mathrm{kg/m^2}$. Meanwhile, counterfactual procedural eligibility under non-Schnur statutory chest criteria (hospital perioperative threshold BMI $< 40.0\,\mathrm{kg/m^2}$, $W_{\text{clin}} \geq 150\,\mathrm{g}$) across the identical anatomic cohort was 92.37%.

As shown in Figure 1C, this policy divergence produced a population-weighted **Asymmetric Access Gap (ΔP_{access}) of** 46.34 **percentage points against Regime A and** 57.80 **percentage points against Regime B**, peaking at 66.86% (vs. Regime A) and 100.00% (vs. Regime B) in Class I obesity (BMI 30.0–34.9 kg/m^2; note that in BMI $\geq$ 35.0 kg/m^2, ΔP_{access} moderates to 49.77% and 59.51% only because hospital anesthesia review also defers cases with BMI $\geq$ 40.0 kg/m^2). Implementing Regime D increased somatic approval among severely impaired non-dysphoric patients to 97.58% (reducing severe-pain somatic denial to 2.42%, attributable solely to incomplete multi-domain chart documentation) and reduced the population-weighted asymmetric access gap ΔP_{access} to 0.00% (100.0% elimination of aggregate cross-silo distortion pressure, with stratum-level residual gaps $\leq$ 2.63%).

3.4 Decision-Curve Net Clinical Utility

Figure 1D displays Vickers–Elkin decision curves across clinical threshold probabilities $p_t \in [0.05, 0.75]$. At a canonical surgical referral threshold of $p_t = 0.25$ (willingness to operate on up to four well-documented candidates to prevent three false-negative denials of refractory chronic pain), population-weighted Net Clinical Benefit was 0.7076 under Regime D compared with 0.3271 under Regime A and 0.2480 under Regime B (0.7085 for indiscriminate Approve-All). At higher diagnostic stringency ($p_t = 0.50$, equal weighting of false-positive surgery and false-negative pain denial), Approve-All deteriorated sharply to 0.5627, whereas Regime D sustained a Net Clinical Benefit of 0.7072—more than double Regime A (0.3268) and nearly triple Regime B (0.2477). Across every threshold $p_t \geq 0.25$, Regime D strictly dominated all anthropometric and treat-all comparators.

4 Discussion

4.1 Psychometric Invalidity of `BSA`-Indexed Gram Quotas and `BMI` Bans

Our Monte Carlo policy audit demonstrates mathematically and empirically why the Schnur 22nd-percentile curve fails as a medical-necessity classifier. Because Schnur's curve increases exponentially across `BSA` ($\beta_{\text{Schnur}} = 1.7687\,\text{m}^{-2}$) while anatomically safe, torque-relieving breast parenchymal excision scales sub-exponentially ($\beta_{\text{clin}} = 0.5800\,\text{m}^{-2}$), the Schnur table conflates systemic body habitus and standing stature with breast-specific orthopaedic pathology. Requiring 732.8 g to 1,058.7 g per breast in taller women with BMI $\geq$ 30 kg/m^2 forces surgeons into an untenable clinical dilemma: either decline prior authorization for patients in genuine chronic pain, or over-resect parenchymal tissue beyond vascular pedicle safety margins merely to satisfy an insurer's pathology-weight audit [3, 13, 19].

Equally unsupported is the compounding of Schnur quotas with blanket outpatient BMI $<$ 30 kg/m^2 exclusions (Regime B), which denied 68.24% of all somatic macromastia candidates and 100.00% of candidates with Class I or higher obesity while adding virtually zero diagnostic specificity (+0.02%). Multicenter outcomes analyses of 2,492 reduction mammaplasty patients by Gust et al. [18] and the 2022 `ASPS` Clinical Practice Guideline [19] demonstrate that while minor wound-healing events increase gradually at extreme `BMI` tiers, major surgical morbidity remains low across BMI 30–39 kg/m^2 and symptomatic relief is at least as large in higher-`BMI` patients whose chronic cervical-thoracic strain makes unassisted weight-loss exercise physically painful [15, 11]. Where ambulatory surgical centers maintain facility anesthesia limits (commonly BMI $<$ 35 or $<$ 40 kg/m^2), clinical triage belongs in hospital-outpatient perioperative anesthesia assessment rather than categorical insurance benefit denial.

4.2 Resolving Diagnostic Siloing Without Compromising Statutory Protections

A central contribution of this audit is formalizing the **Asymmetric Access Gap (ΔP_{access})** created when modern statutory reforms eliminate Schnur gram tables in gender-affirming chest policies (such as `Minn. Stat. § 62Q.585` [25]) while payers leave 1991 Schnur and `BMI` gates intact on somatic `CPT 19318`. Importantly, the policy defect does *not* lie in statutory `GAMST` protections—which properly align gender-affirming surgical review with contemporary clinical standards—but rather in payers' failure to modernize somatic macromastia policies in parallel. When 65.43% of non-dysphoric women in severe daily physical pain ($Q_{\text{phys}} \leq 50$) are blocked from `CPT 19318` under Regime B (53.97% under Regime A) while the identical surgical procedure carries no gram quota in an adjacent coverage silo, health systems manufacture unnecessary administrative friction, patient distress, and iatrogenic pressure toward diagnostic misrouting. Conversely, in Canadian provinces where public schedules impose strict gram or clinic `BMI` ceilings alongside multi-year surgical queues without clear non-flat reduction parity, both somatic and gender-diverse candidates suffer protracted untreated disability [17].

Adopting Regime D resolves this asymmetry upward rather than downward. By anchoring somatic `CPT 19318` medical necessity to a Rasch-calibrated physical well-being deficit ($Q_{\text{phys}} \leq 62.0$, representing $\geq 2 \times$ MCID below general-population norms [8, 12]) paired with ≥ 2 clinician-documented chronic somatic domains refractory to conservative care, payers retain 99.73% specificity against purely cosmetic claims (FDR = 0.08%) while lifting severe-pain somatic approval to 97.58% and eliminating the aggregate dual-silo access gap ($\Delta P_{\text{access}} = 0.00\%$). Patients seeking somatic pain relief never face pressure to assert an identity diagnosis they do not hold, while transgender and non-binary patients retain full statutory protection under `GAMST` pathways. For clinics without licensed proprietary patient-reported outcome software, public-domain Rasch-equated or structured numeric functional pain logs paired with the four canonical somatic exam domains provide an immediately deployable administrative bridge.

4.3 Limitations

Three methodological limitations warrant note. First, although marginal distributions and correlations were calibrated to landmark prospective and normative datasets [1, 2, 14, 9, 8, 10, 12], real-world payer contracts exhibit local heterogeneity, including alternate flat gram caps (e.g., covering $\geq 500\,\text{g}$ or $\geq 1,000\,\text{g}$ regardless of higher `BSA`) and federal `ERISA` preemption among self-funded employer plans. Second, our simulation models preoperative Rasch scores without intentional respondent gaming; pairing self-reported Rasch impairment with objective physical exam signs ($K_{\text{dom}} \geq 2$ documented over ≥ 6 months plus conservative management) specifically mitigates single-instrument response bias. Third, prospective multi-payer pilot implementation studies are needed to empirically validate administrative claims savings from reduced peer-to-peer appeals and avoided chronic musculoskeletal analgesia utilization.

5 Conclusions

Thirty-five years after its publication as a descriptive survey curve, the Schnur 22nd-percentile `BSA` sliding scale—particularly when compounded by categorical BMI $< 30\,\text{kg/m}^2$ exclusions—wrongly denies medically necessary care to 58.12%–68.24% of true somatic macromastia patients, penalizes taller women even at healthy body weights, and generates a 46.34–57.80 percentage-point dual-silo access gap for patients in severe physical pain. Replacing arbitrary anthropometric cutoffs with Rasch-calibrated physical well-being thresholds and documented multisystem somatic pathology

achieves $>$ 90% sensitivity, $>$ 99.7% specificity, and superior net clinical benefit across North American health systems.

Reproducibility and Data Availability

All simulation code (`01_simdesign_schnur_vs_rasch.R`), read-only anthropometric lookup functions (`toolkit/schnur_bsa_calculator.R`), condition-level outputs (`policy_condition_summary.csv`), and aggregate verification metrics (`policy_simulation_metrics.json`) are deterministically reproducible from `set.seed(20261102)` in `R` version 4.4.2. Zero private or human-subject datasets were used.

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